



Lecture 1: Course Kickoff

Business Applications (Betriebswirtschaftliche Anwendungen)

Prof. Dr. Robin Gubela
SoSe 2026



Hochschule für Technik
und Wirtschaft Berlin

University of Applied Sciences



Agenda

1

Course Logistics

2

Basics of Processes

3

Basics of Application Systems

Prof. Dr. Robin Gubela

Professorship for Business Applications and Business Processes



Professional Experience

- Federal Ministry of Transport – Data Science Specialist
- Deutsche Kreditbank – Senior Portfolio Manager
- Boston Consulting Group Platinion – Strategy Consultant

Academic Background

- HWR Berlin – Research Associate at EMF Institute
- Media University, HU Berlin – Lecturer E-Business & Machine Learning
- HU Berlin – Ph.D. in Information Systems
- Vrije Universiteit Amsterdam – Exchange Term Master Consulting
- HU Berlin – M.Sc. Information Systems
- FU Berlin – B.Sc. Business Administration



Bundesministerium
für Verkehr



BCG
PLATINION



Hochschule für
Wirtschaft und Recht Berlin
Berlin School of Economics and Law

DKB
Deutsche Kreditbank AG

HUMBOLDT-UNIVERSITÄT
ZU BERLIN



Media
University
of Applied Sciences

VU
JSSS

VRIJE
UNIVERSITEIT
AMSTERDAM

Freie Universität



Berlin

htw

Warm up

- **Did you complete your bachelor's degree in Berlin?**
Window – yes | Wall – no
- **Where do you live in Berlin?**
Board – North | Behind you – South | To your left – West | To your right – East
- **Do you have work experience?**
Window – yes | Wall – no
- **Do you have a general idea of the topics covered in this course?**
Window – yes | Wall – only vaguely | Board – no
- **Can you code?**
Window – very well | Wall – sort of | Board – not at all

Course content (preliminary)

1	Course Kickoff	7	AI agent ecosystem & Retro
2	Introduction to ERP systems	8	AI agents in business workflows
3	SAP ERP	9	Agentic ERP in S/4 HANA
4	Principles of ERP systems	10	OpenClaw & Test preparation
5	Modern ERP concepts	11	E-Test
6	Fundamentals of AI agents		

Today

Join on
[slido.com](https://www.slido.com)
with # 9908

Your expectations

Opportunity to share your expectations – I will do my best to address them



What are your expectations for the module Business Applications?



What expectations do you have for me as the module instructor?

As part of a retrospective (at about semester halftime), you tell me your learning progress and how I can further support you on your learning path

Student input
April 8, 2026

Your expectations (1/2)

What are your expectations for the module Business Applications?

Anonymous

heard good things from the last semester, so i expect
to be as happy as they were

👍💡 10

Anonymous

have openclaw take my e-test would be fun

🔥👍 6

Anonymous

how to apply ai agents to improve workflows maybe?

👍💡 5

Anonymous

discuss ai agents like openclaw in a critical manner

🔥 4

Anonymous

Dangers of giving ai Access to your erp

🔥👍 2

Anonymous

I'd like to learn more about using AI agents effectively

👍 2

Anonymous

for now not much

👍👍 2

Anonymous

Learn more about AI and openclaw

👍 1

Anonymous

OpenClaw discussions

🔥 1

Anonymous

To learn about AI, ERP and the integration of them

🔥 1

Anonymous

Learn more about AI agents and how to improve workflows

😊 0

Anonymous

Learn about how ERP systems work in the background

😊 0

Anonymous

to learn SAP

😊 0

Anonymous

learn how companies use erp

😊 0

Anonymous

How to integrate Software oder Smart systems for
Companies

😊 0

Anonymous

to learn more about sap and how to use and maybe setup
ai agents

😊 0

Anonymous

How to Integrate ai into erp Systems

😊 0

Anonymous

To learn sap

😊 0

Student input
April 8, 2026

Your expectations (2/2)

What expectations do you have for me as the module instructor? ✨

Clear communication (with clear deadlines, expectations, what to learn)

👍💡 5

The best way I learn is through real-life examples or company scenarios 😊

👍🔥 3

showing openclaw live and in general more practical lessons

🔥👍 2

A friendly, approachable, and supportive professor who explains things clearly and doesn't make the module unnecessarily complicated.

🔥👍 2

Give real experience / advices from your jobs

👍 2

a well structured moodle class

👍 2

Be more practical oriented

🔥 1

To not just read the slides

🍌 1

Be funny. Tell interesting stories

🔥 1

onlinesessions ?

😐 0

I expect a clear structure, understandable explanations, fair treatment, and helpful feedback

😐 0

to start from the basics, so everybody understands

😐 0

Ground rules for the course

- **Respect for others**
e.g., listen to others, let others finish speaking, do not disrupt class, etc.
- **Taking photos in class is not permitted**
Slides will be made available on Moodle anyway
- **Punctuality is expected**
Starting 5 minutes after the lecture begins, **only three additional latecomers will be admitted to the lecture**, no further admissions after that

Grading

Module grade is calculated as the average of the scores from the lecture (max. 50 points) and the tutorial (max. 50 points)

- **Lecture**

- ERP presentation (30 points)
- E-Learning Agentic AI (10 points)
- In-class test (10 points)

- **Tutorial**

- Prototype development (30 points)
- Prototype presentation (20 points)



Submission deadlines

Strict adherence to deadlines – no exceptions

- **Late submissions will not be accepted** – You will receive 0 points, regardless of the reason for the delay.
- **Technical problems do not change this** – Temporary Moodle outages, accidentally uploading a wrong file, forgotten login credentials, or other technical issues will not be taken into account.
- **Only Moodle submissions will be graded** – Submissions via e-mail, for example, will not be accepted. Therefore, use Moodle for all submissions.
- **Tip:** Submit at least one day before the deadline to avoid last-minute problems. Moodle will confirm your successful upload.
- **Why this rule?** Fairness toward all students and realistic preparation for the professional world – application deadlines, project deadlines, etc. are binding.

Only work submitted on time via Moodle will be graded – Missing or late submissions will automatically result in 0 points for the respective assignment

In-class questions

- Are **explicitly encouraged**
- Targeted questions can help you **close knowledge gaps**
- There are **no “stupid” questions**
- During lectures, you have the opportunity to ask questions **at any time**

Out-of-class questions

I supervise many hundreds of students at HTW Berlin. If you have questions outside of class, please follow this **order** of steps:

1. Check whether you can answer the question yourself (e.g., using course materials/Moodle)
2. Ask your fellow students
3. Ask during the next class session
4. Ask directly before or after the next class session

Only after these steps have been exhausted – or in case of a hardship situation – send me an e-mail from your HTW e-mail address





Agenda

1

Course Logistics

2

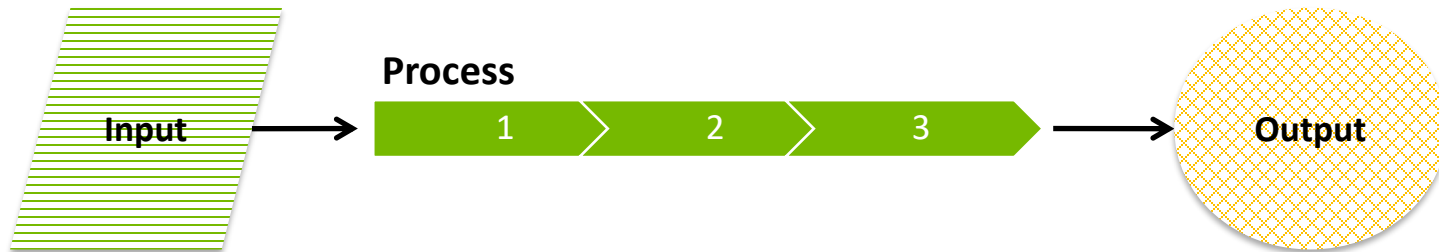
Basics of Processes

3

Basics of Application Systems

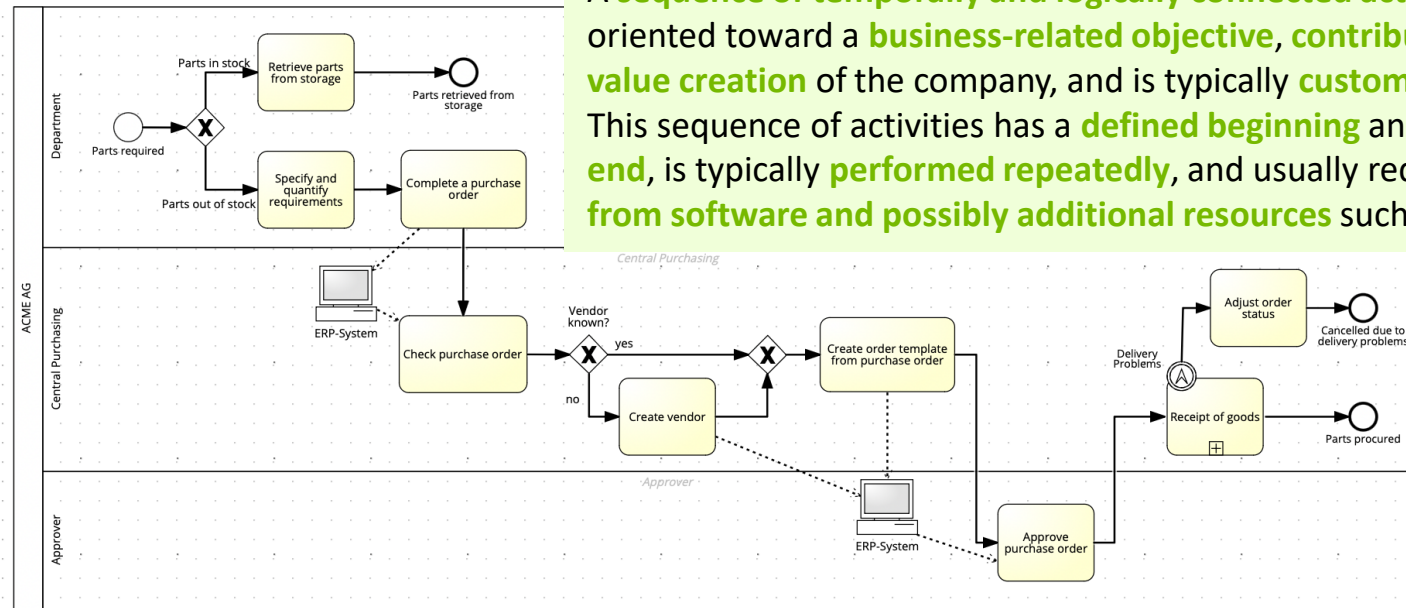
Definition of process

- A sequence of logical **individual functions** with a **connection** between them
- By executing functions, one or more **input factors** are transformed into one or more **output factors**.
- What is being processed is **matter and/or information**.



Definition of business process

A **sequence of temporally and logically connected activities** that is oriented toward a **business-related objective**, contributes to the **value creation** of the company, and is typically **customer-oriented**. This sequence of activities has a **defined beginning** and a **defined end**, is typically **performed repeatedly**, and usually requires **support from software and possibly additional resources** such as machinery.



Types of business processes

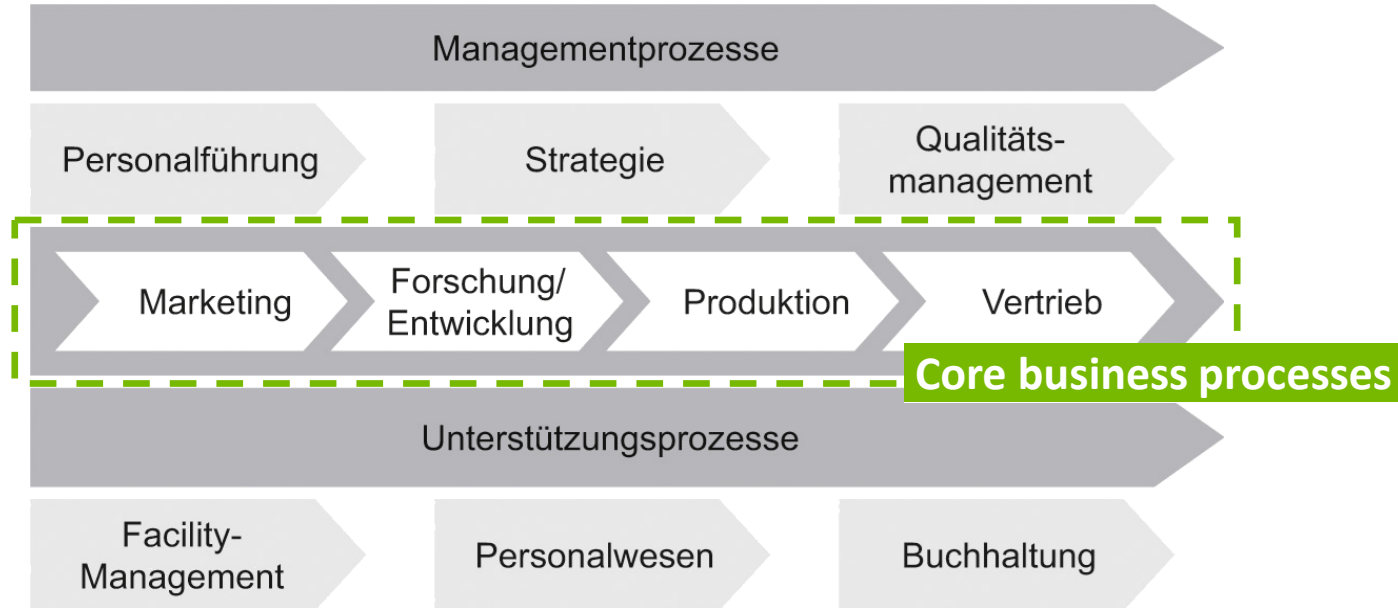
Core business process

A process that directly contributes to value creation

Support process

A process that does not directly contribute to value creation but is still necessary for business operations

Core business processes



Business process management

Goal

Business process management (BPM) is the totality of all activities a company undertakes to **improve its processes** (e.g., greater efficiency and effectiveness).

Tasks

Identification, planning,
and organization of
business processes

Optimization of
business processes

Documentation of
business processes

Controlling and
monitoring of business
process instances

Tools

Modeling

Simulation

Workflow management

Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes





Agenda

1

Course Logistics

2

Basics of Processes

3

Basics of Application Systems

Fundamentals of application systems



- Dedicated systems for functional areas of a company
- Partly running on their own hardware, which manages the necessary data independently
- **Notable:** Systems often do not communicate directly with each other

Problems with business application systems

- **Limited data exchange**, e.g., sales can't view current inventory during sales process
- Large number of **interface programs** required
- **Data redundancy**: Same data must be maintained in multiple functions
- **Data inconsistencies**, e.g., differing attributes, temporally uncoordinated data changes, etc.
- With **media breaks**: Repeated data entry is required, which often causes significant time expenditure and increased error potential

➡ Negative impact on business efficiency

Integration of application systems

Fundamentals

- integrare = to restore
- Here: Merging of previously separate application systems
- IT-based provisioning of interlinked business areas from a unified system
- A new comprehensive unit composed of complementary components
- Idea: More value than the mere sum of its parts

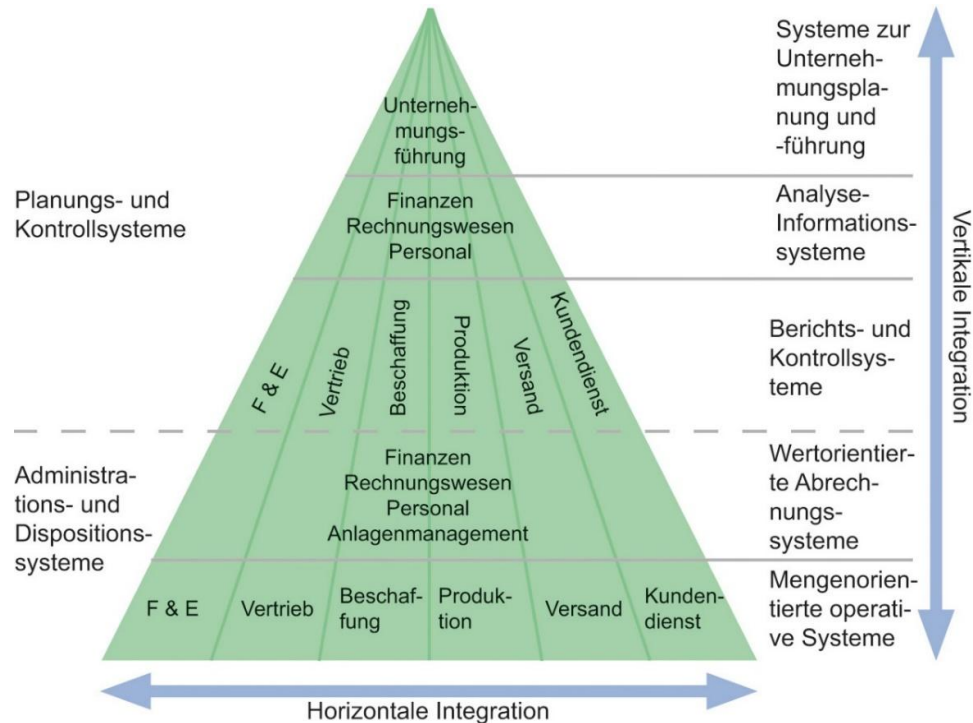
Characteristics

I. Integration direction

II. Integration scope

III. Integration objects

I. Integration direction



Vertical Integration

- Data supply for planning and control systems from operational systems
- Consolidation of data into information
- Example: Overview of all quotes

Horizontal Integration

- Linking the systems along the business value chain
- Involved subsystems, e.g., from sales, materials management, human resources

II. Integration scope

Inter-Company Integration

Coordination of IT systems of at least two companies so that data (e.g., orders) can be exchanged digitally between them

Intra-Company Integration

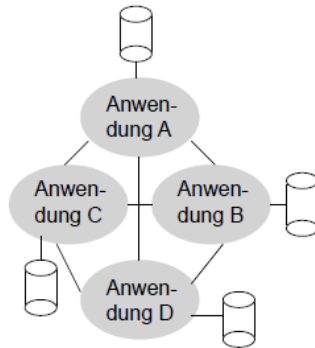
Cross-departmental and cross-process connection within a company

Example: ERP system

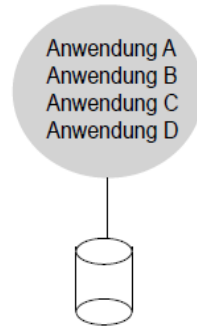
III. Integration objects

Data Integration

- Shared data usage by software modules of a system
- Example: Customer master data used by the sales module and accounting module



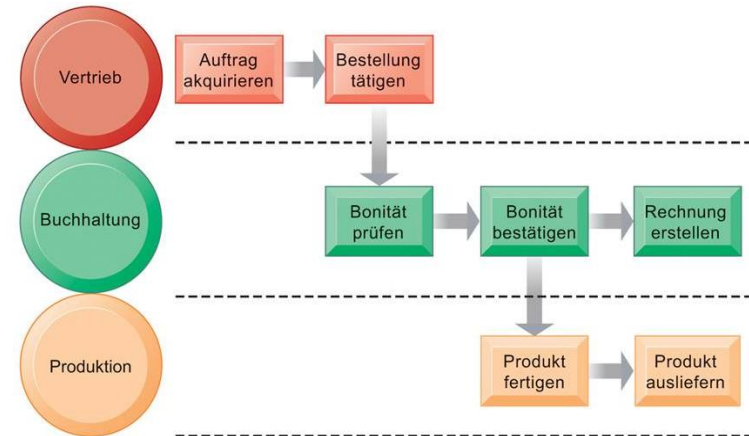
Applications with separate data storage



Applications with shared data storage

Process Integration

- Multiple software modules support a cross-departmental business process
- Example customer order processing: Handled via the integrated SAP modules SD, PP, and FI



Evaluation of application system integration

Advantages

- Reduction of manual data entry effort
- Reduction of data entry errors
- Reduction of data inconsistencies
- Faster error detection through multiple use of data
- Fixed, automated workflows increase process quality
- Shorter processing times, opportunity to optimize workflows

Challenges

- Chain reaction in case of errors
- High testing and maintenance effort due to complexity
- Qualified system planners may be hard to find
- High onboarding effort for system administrators
- Process adjustments may be necessary
- Long implementation and investment timelines

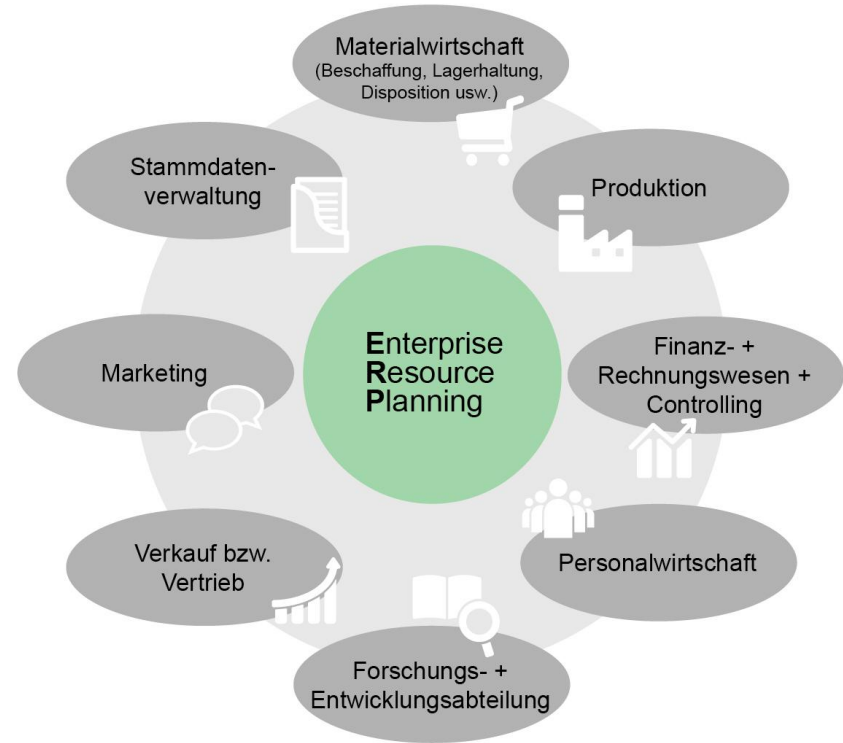
ERP systems

Enterprise Resource Planning (ERP) systems

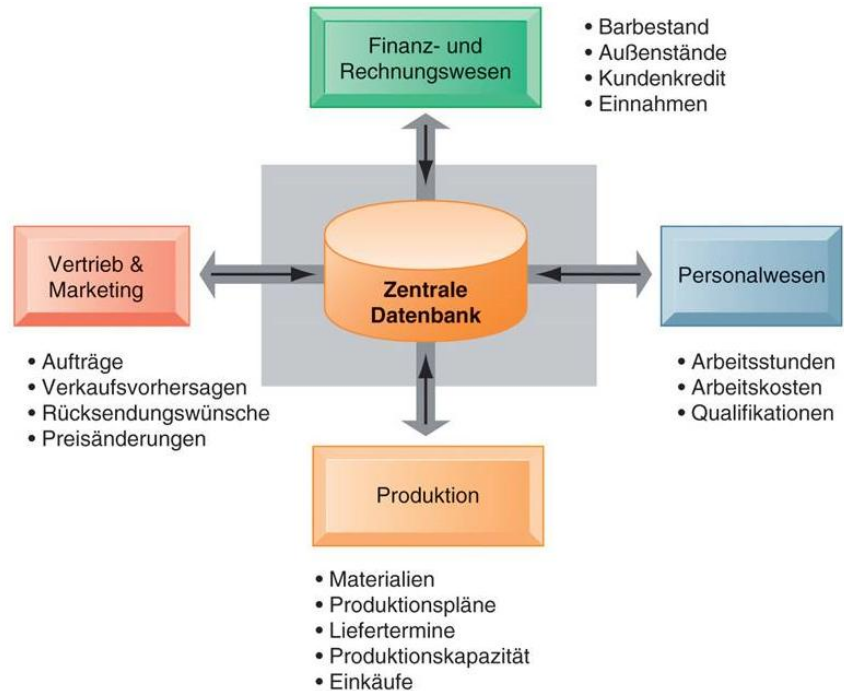
are extensive standard business application software packages that support nearly all typical business processes of a company

Properties

- Integrated software modules for business functions (see figure on the right)
- A central database
- Real-time availability
- Modular architecture



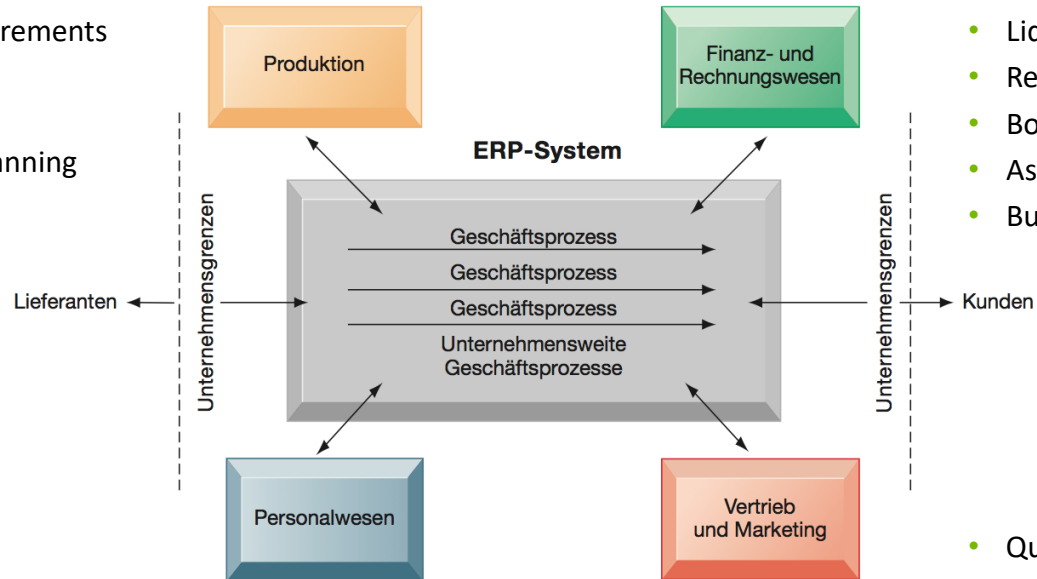
Central database



- Storage of data from various departments in **one central database**
- Departments work with **the same data**
- Data **created in other departments** can be used, e.g., shipping department can view customer data created by sales
- **Application example SAP S/4HANA**

Functions of ERP systems

- Inventory management
- Material requirements planning
- Purchasing
- Production planning



- Financial planning
- Liquidity management
- Receivables and payables
- Bookkeeping
- Asset accounting
- Budget planning/monitoring

- Payroll
- Bonuses and premiums

- Quote management
- Order entry
- Sales analyses

References

Abts, D. und W. Müller (2017): Grundkurs Wirtschaftsinformatik. 9. Auflage, Springer Vieweg Verlag.

Gadatsch, A. (2020): Grundkurs Geschäftsprozess-Management. Analyse, Modellierung, Optimierung und Controlling von Prozessen. 9. Auflage, Springer Vieweg Verlag.

Gronau, N. (2021): ERP-Systeme: Architektur, Management und Funktionen des Enterprise Resource Planning. 4. Auflage, De Gruyter Oldenbourg Verlag.

Hansen, R. H., Mendling, J., Neumann, G. (2019): Wirtschaftsinformatik. 12. Auflage, De Gruyter Oldenbourg Verlag.

Krcmar, H. (2015): Informationsmanagement. 6. Auflage, Springer Gabler Verlag.

Laudon, Laudon und Schoder (2015): Wirtschaftsinformatik – Eine Einführung, 3. Auflage, Pearson Verlag, 2015

Leimeister, J. M. (2021): Einführung in die Wirtschaftsinformatik. 13. Auflage, Springer Gabler Verlag.

Weber, Gabriel, Lux und Menke (2022): Basiswissen Wirtschaftsinformatik, 4. Aufl., Springer Vieweg, 2022

Thank you for your attention



Prof. Dr. Robin Gubela

Business Applications and Business Processes

Department 4 – Information Systems

robin.gubela@htw-berlin.de

www.htw-berlin.de



Hochschule für Technik
und Wirtschaft Berlin

University of Applied Sciences



**Hochschule für Technik
und Wirtschaft Berlin**

University of Applied Sciences

www.htw-berlin.de